

OneModel Traffic Light Annotation Specification

Red-Green Light Annotation Spec — English Translation

Version History

Version	Date	Changes
V-E2E-2.0	2025-03-14	Initial release
V-E2E-2.1	2025-04-17	Updated rules and criteria

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Last Modified: April 17, 2026

1. Annotation Overview

This section defines the acceptance criteria for traffic light annotations, covering lamp boxes, bulb points, and passability quick judgments.

1.2 Abnormal Image Judgment

Images that do not meet quality standards should be classified as ignorable or invalid per the project criteria.

1.3 Acceptance Criteria

Category	Focus	Scale	Lamp Box	Bulb Point
Miss			• If bulb points are annotated, lamp box must also be annotated • Telephoto (HD): short side > 10px • Wide-angle: short side > 5px	If a lamp box is annotated, all bulb points must be annotated
Extra			• Does not meet pixel requirements • Lamp box never lit during entire sequence • Lamp box with no visible bulbs	Lamp boxes where all short sides < 10px
Fit			Fitting principle: error < 3px	• Fitting principle: error < 3px • Bulb point placed on occluding object

Passability Quick Judgment

When the lamp box short side is larger than 5 pixels, judge the label according to the rules.

Accuracy Requirement: Acceptance elements + quick judgment accuracy must exceed 98%. All errors found during acceptance must be corrected (including attribute and type errors).

2. TL Rules — Bounding Box Annotation

This section defines the annotation objects, attributes, and subcategories for traffic light labeling.

Annotation Objects and Attributes

Category	Object	Attribute	Subcategories
Lamp Box		Type	Pedestrian light, Motor vehicle light, Bicycle light, Other, Unknown
		Occlusion	No occlusion, 1%~50%, 50%~100%
Bulb Point		Bulb ID	Numeric ID
		Shape	Circle, Countdown, Up arrow, Left arrow, Right arrow, Bicycle, Pedestrian, U-turn arrow, Left-turn U-turn, Straight+Left, Straight+Right, Down arrow, No entry, Other arrow, Blur/Uncertain, Chinese text, Other
		Color	Red, Yellow, Green, Black, Unknown
		Flashing	Yes / No
		Countdown	Specific value. For non-countdown lights, fill the countdown value of the corresponding countdown light. If none, write -1.

2.1 Lamp Box

2.1.1 Annotation Requirements

Precision Requirement: Annotate close to the edges of the lamp box. If there is a white border, annotate along the white border edge. Overlapping 2D boxes caused by actual positioning are acceptable.

[IMAGE PLACEHOLDER] — *Lamp box edge fitting examples (1280×1092px, 1280×752px)*

- **Special lamp box annotation examples:**
- Lamp with text signboard: annotate the sign and the lamp together as one box
- Lamp and text are not integral: annotate only the lamp
- Broken/damaged areas: do not need to be fully enclosed in the box

[IMAGE PLACEHOLDER] — *Special lamp box examples (419×351px)*

2.1.2 Lamp Box Categories

2.1.2.1 Motor Vehicle Light ()

Standard traffic lights controlling motor vehicle flow. Includes standard 3-light (red/yellow/green), arrow-direction lights, and countdown displays.

2.1.2.2 Pedestrian Light

Traffic lights for pedestrian crossings. Typically show walking person (green/walk) and standing person (red/stop) symbols.

2.1.2.3 Bicycle Light

Traffic lights designated for bicycle lanes. Similar to motor vehicle lights but specifically for bicycle traffic flow.

2.1.2.4 Other

Lamp boxes that do not clearly fit into the above categories, including auxiliary or special-purpose traffic signals.

2.1.2.5 Unknown

Lamp boxes that cannot be determined due to image quality, occlusion, or distance. Auxiliary road traffic lights without clear bicycle/pedestrian markings that cannot be distinguished by scene context should default to motor vehicle light.

2.2 Bulb Point

Bulb points are individual light elements within a lamp box. Each visible illuminated bulb must be annotated as a separate point.

2.2.1 Annotation Requirements

- Every lamp box that is annotated must have all its visible bulb points annotated
- Bulb points must be placed on the center of the visible bulb, not on occluding objects
- Fitting principle: error < 3px
- For lamp boxes where all short sides are < 10 pixels, bulb points should not be annotated (over-annotation)

2.2.2 Bulb Point Attributes

Attribute	Values / Description
Bulb ID	Numeric ID — sequential number assigned to each bulb point within a lamp box
Shape	Circle, Countdown, Up arrow, Left arrow, Right arrow, Bicycle, Pedestrian, U-turn arrow, Left-turn U-turn, Straight+Left, Straight+Right, Down arrow, No entry, Other arrow, Blur/Uncertain, Chinese text, Other
Color	Red, Yellow, Green, Black, Unknown
Flashing	Yes / No
Countdown	Specific numeric value. For non-countdown lights, fill the countdown value of the corresponding countdown light. If none exists, write -1.

3. Passability Quick Judgment Rules

Passability determination applies when the lamp box short side exceeds 5 pixels. Below this threshold, passability is not judged.

3.1 Passability Judgment Criteria

- Judge passability based on the color and state of the traffic light
- Red = Stop (not passable), Green = Go (passable), Yellow = Caution
- Flashing lights follow the same color rules
- Countdown displays: use the color of the associated signal
- When a lamp has multiple lights (e.g., left arrow green + straight green), each direction is judged separately

3.1.2 Existence Attribute

For auxiliary road traffic lights where no bicycle/pedestrian symbol is present and the scene cannot determine the type, default to motor vehicle light classification.

[IMAGE PLACEHOLDER] — Auxiliary road light example (from Feishu wiki)

Appendix A: Image Reference Notes

The following images were present in the original Feishu wiki but could not be extracted due to session-bound blob URLs. They should be manually added from the source document.

#	Description	Original Size	Section	Action
1	Lamp box edge fitting — precision examples	1280×1092	2.1.1	Manual capture
2	Lamp box edge fitting — wide-angle	1280×752	2.1.1	Manual capture
3	Special lamp box examples — text signboard	419×351	2.1.1	Manual capture
4	Lamp with text: annotate together	~300×200	2.1.1	Manual capture
5	Lamp with separate text: annotate only the lamp	~300×200	2.1.1	Manual capture
6	Broken/damaged lamp: box does not need full coverage	~300×200	2.1.1	Manual capture

Note: Sections 1.2 (Lamp Box Category Detail with illustrations), 2.3 (Bulb Point Classification with multi-lamp-box scenarios), and 3.x (Passability sub-rules with countdown binding examples) were listed in the source document's table of contents but their full content had lazy-loading issues during extraction. The attribute tables above contain the complete classification values. For detailed illustration examples and multi-lamp-box scenarios, refer to the original Feishu wiki.